

Song-Ze Yu

E-mail: vaclis@berkeley.edu | Personal Page: <https://vaclis.net/> | GitHub: github.com/vaclisinc

Research Interests

My research focuses on grounded **music understanding** in large audio-language models and **controllability for musicians** in music and audio effects generation. My long-term goal is to bring music models toward **artificial music intelligence**, offering musicians new interfaces to directly discuss, collaborate, and interact with machines.

Education

University of California, Berkeley Aug. 2025 - May 2026
Visiting Student in Computer Science Advised by Prof. Trevor Darrell and Carmine-Emanuele Cella
• GPA: 3.9 / 4.0

National Tsing Hua University Sep. 2023 - Jun. 2026
B.S. in Computer Science Advised by Prof. Liu Yi-Wen
• Overall GPA: 3.91 / 4.30

National Cheng Kung University Sep. 2022 - Jun. 2023
B.S. in Computer Science and Information Engineering
• Overall GPA: 4.16 / 4.30; Rank: 5 / 113 (Top 5%)

Publications (* indicates equal contribution)

Music Understanding in Audio-Language Models

- [1] [PitchBench: Measuring Pitch Hearing in Audio-Language Models](#) **NeurIPS 2026**
Milan Liessens Dujardin*, **Song-Ze Yu***, Craver Corbyn Thomas-Smith, David M. Chan, Karina Nguyen Evaluations & Datasets
- [2] [MuSP-Bench: Advanced Multimodal Benchmarking of Music Understanding across Score and Performance](#) **ISMIR 2026**
Milan Liessens Dujardin, **Song-Ze Yu**, Kevin Miao Late-Breaking Demo
- [3] [Unifying Score and Performance for Fine-Grained Music Understanding in Audio-Language Models](#) *Under review*
Milan Liessens Dujardin, **Song-Ze Yu**, Kevin Miao
- [4] Diagnosing Music Understanding in Audio-Language Models: Encoder-Decoder Probing and Task Adaptation *Under review*
Jun-You Wang, **Song-Ze Yu**, Xuanjun Chen, Chih-Kai Yang, Ke-Han Lu, Sung-Feng Huang
- [5] UniFront: Revisiting Time-Frequency Frontends for Unified Music and Speech Modeling *Under review*
Song-Ze Yu, David M. Chan

Controllability for musicians

- [6] [InstructFX2FX: A Multi-Turn Text-to-Effect System for Sequential Audio Effect Refinement](#) **DAFx 2026**
Song-Ze Yu, Milan Liessens Dujardin, Yuxuan Cai, Wantong Zhang, Brian Cruz, Jeremy Wagner, Carmine-Emanuele Cella Demonstration
- [7] MIDI-REPA: Improving Text-to-Music Generation with Symbolic Representation Alignment *Under review*
Song-Ze Yu, Fang-Duo Tsai, Yi-Hsuan Yang

HCI & human simulation

- [8] [Anamnesis: An Open-Source Platform for Large-Scale Backstory-Conditioned Survey Simulation](#) **EMNLP 2026**
Song-Ze Yu, Joseph Suh, Serina Chang, David M. Chan System Demonstration

Research Experience

Music and AI Lab, National Taiwan University Jun. 2026 - Present
Research Assistant Advised by Prof. Yi-Hsuan Yang

- Leading the end-to-end training of a **music LALM**, with collaborators from NVIDIA, BAIR, Queen Mary University of London, and Prof. Hung-yi Lee's group.
- Researching symbolic representation alignment for text-to-music generation and representation learning in audio-language models, leading to **MIDI-REPA** and **encoder-decoder probing studies**.

Berkeley AI Research (BAIR) Lab, UC Berkeley Feb. 2026 - Present
Researcher Advised by David M. Chan and Prof. Trevor Darrell

- Revisiting the ubiquitous 25 ms log-Mel frontend used across general audio encoders and models under conflicting speech-music resolution demands, leading to **UniFront**.
- Lead-developed **PitchBench** and **Anamnesis**, spanning the basics of music understanding in audio-language models and large-scale human simulation.

Center for New Music and Audio Technologies Lab, UC Berkeley

Undergraduate Research Lead

Sep. 2025 - Jun. 2026

Advised by Prof. Carmine-Emanuele Cella

- Led the research direction on interactive, musician-centered audio effect modeling and editing, extending existing single-shot toward iterative refinement and leading to **InstructFX2FX**.

AHG Music Lab, National Tsing Hua University

Undergraduate Researcher

Jul. 2024 - May 2025

Advised by Prof. Yi-Wen Liu

- Developed the **VTR audio-to-EQ model** for single-shot tone replication as my undergraduate capstone project, later deployed as a VST plugin at Positive Grid.

Selected Engineering Experience

BerkeleyTime

Machine Learning Pod Lead

Sep. 2025 - May 2026

Spring 2026

- Led a **10-member ML Pod** across semantic search, recommendation system, and enrollment prediction, coordinating researcher-engineer pairs and designers from model development through deployment.
- Led development of a **new Explore page**, integrating course recommendations into BerkeleyTime's product experience.

Machine Learning Engineer, Catalog Pod

Fall 2025

- Built and deployed **BGE + FAISS semantic search** for **122K+ monthly users** using FastAPI.
- Reduced GraphQL first-load latency from **25s to 1s** with pagination and Redis caching; shipped and fixed **20+ PRs**.

Positive Grid

Machine Learning Intern

Jun. 2025 - Aug. 2025

Taipei, Taiwan

- Served as Scrum Master for **Amp-AI-SaaS** and designed the **Multi-Agent Orchestration Harness** for ongoing software-engineering workflows at Positive Grid.
- Integrated my capstone **VTR audio-to-EQ model** into a functional **JUCE VST plugin**, integrating PyTorch inference for real-time audio processing in DAW.

Awards & Leadership

Meichu Hackathon — 3rd Place, NXP Group

Sep. 2026

- Built **GodEyes**, a physical AI system using a **world model** to turn crime-scene photos into explorable 3D worlds.

Claude × a16z × Berkeley M.E.T. Makeathon — Best Design

Oct. 2025

- Built Haven in five hours, a mobile app that personalizes home-design recommendations from room scans.

Meichu Hackathon — 2nd Place, Google Group

Oct. 2024

- Built **Travelity**, an AI travel planner using interests, personality, and real-time map and search data.

Taipei Hackathon — 3rd Place

Sep. 2024

- Built **ParkFlow**, a WebView microservice for real-time parking information, navigation, and notifications; selected for integration into Taipei's official city app (**3M+ downloads**) and awarded NT\$100K.
- Produced a **hackathon vlog** (**1K+ views**); **invited by the Taipei City Government** to share the experience a year later.

SITCON Hackathon — Group Award Winner

Jul. 2024

- Built **ccDiary**, an AI-assisted mental-health journal with emotion analysis and anonymous social features.

National Cheng Kung University — Campus Speaker

Aug. 2023

- Delivered an **invited talk** for incoming students on writing music as a diary, then turned audience-submitted phrases and sentences on Slido into improvised songs performed live.

Selected Music Experience

Pianist, Composer & Producer

Music CV | YouTube (7K subscribers)

- Performed **24 solo piano recitals** across Greater China and appeared on **70+ TV shows and New Year galas (春晚)** between 2009 and 2016.
- Shared the stage with **Jay Chou (周杰倫)** **three times** and performed eight-hands piano at Taipei Arena with Bai-An (白安) and Cosmos People (宇宙人) as a guest at Yan Jue's (嚴爵) concert.
- Wrote **20+ original songs with 20K+ views** and continues to compose, perform, and produce original music.